

Probability and Optimization IA-3

(1) Let

$$f_X(x) = \begin{cases} x + \frac{1}{2}, & 0 < x < 1, \\ 0, & \text{otherwise} \end{cases}$$

be the density function of the random variable X .

Suppose $Y = 2X + 1$. Then:

- (a) Find the cumulative distribution function (CDF) of Y .
- (b) Find the density function of Y .
- (c) Find the density function of $Y = X^2$.

(2) A cricket fan starts with Rs. 200 and places bets of Rs. 100 on each match. For every match, with probability $\frac{2}{3}$, his prediction is correct and he wins Rs. 100; with probability $\frac{1}{3}$, his prediction is wrong and he loses Rs. 100.

He stops betting if his total money becomes either Rs. 0 or Rs. 400.

- (a) Find the probability that he loses all his money after exactly three matches.
- (b) In the long run, find the probability that he eventually loses all his money.